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∴ Simulate Static Routing Configuration using CISCO Packet Tracer.

Aim:-

To simulate a static routing configuration using CISCO Packet Tracer.

Procedure:-

- * Open Cisco Packet Tracer and create the given network topology.
- * Connect 3 routers and PCs using suitable cables.
- * Configure IP addresses for routers and PCs.
- * Enable all router interfaces using no shutdown.
- * Configure static routes on Router 0, Router 1, and Router 2 using ip mobile.
- * Set main and backup router using different Administrative Distance values.
- * Test Connectivity
- * Use tracer to verify the routing path.
- * Disconnect the main link and verify that the backup route works.
- * Thus, a static routing is successfully configured and verified.

Router0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

Router0#show ip interface brief
Interface      IP-Address      OK? Method Status
GigabitEthernet0/0  40.0.0.1        YES manual up
GigabitEthernet0/1  20.0.0.1        YES manual up
Router0#show ip route static
Static Routes: 0 - static, 1 - ICMP, 8 - RIP, 9 - mobile,
               8 - OSPF, 0 - EIGRP, 13 - EIGRP external
Gateway of last resort is not set
S  30.0.0.0/8 via 20.0.0.2
S  20.0.0.0/8 via 10.0.0.2
Router0#

```

Router1

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

Router1#show ip interface brief
Interface      IP-Address      OK? Method Status
GigabitEthernet0/0  40.0.0.2        YES manual up
GigabitEthernet0/1  20.0.0.2        YES manual up
Router1#show ip route static
Static Routes: 0 - static, 1 - ICMP, 8 - RIP, 9 - mobile,
               8 - OSPF, 0 - EIGRP, 13 - EIGRP external
Gateway of last resort is not set
S  30.0.0.0/8 via 20.0.0.1
S  20.0.0.0/8 via 10.0.0.1
Router1#

```

Router2

Physical

Config

CLI

Attributes

IOS Command Line Interface

Router3

Physical

Config

CLI

Attributes

IOS Command Line Interface

PC0

Physical

Config

Desktop

Programming

Attributes

Extended Packet

PC1

Physical

Config

Desktop

Programming

Attributes

Extended Packet

Result ✓

Thus, static routing was successfully configured in Cisco Packet Tracer. Connectivity between all networks was verified successfully using ping.